INCH-POUND
MIL-DTL-62548C(AT)
22 September 2010
SUPERSEDING
MIL-DTL-62548B(AT)
24 August 1998

DETAIL SPECIFICATION

LEAD, IGNITION

This specification is approved for use by the U.S. Army Tank Automotive Research, Development and Engineering Center (TARDEC), Research, Development and Engineering Command (RDECOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an ignition lead that transmits high energy and high tension voltage from an ignition exciter to an igniter plug with energy to ignite and start a gas turbine engine (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document cited to the extent specified herein. Unless

Comments, suggestions, or questions on this document should be addressed to U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to DAMI_STANDARDIZATION@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

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otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

FEDERAL SPECIFICATIONS

A-A-52557	-	Fuel Oil, Diesel

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-6083	-	Hydraulic Fluid, Petroleum Base, for Preservation and Operation
MIL-PRF-7808	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-DTL-16884	-	Fuel Oil, Naval Distillate
MIL-PRF-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number 0-156

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	U.S. Military Property, Identification Marking of
MIL-STD-461	-	Control of Electromagnetic Interference Emissions and Susceptibility, Requirements for (see 4.2.1)
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines (see 4.2.1)

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from Document Automation and Production Service, Building 4/D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or website: https://assist.daps.dla.mil/quicksearch/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract (see 6.2).

DRAWINGS

12286829	-	Lead, Ignition

(Copies of these documents are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

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2.3 Non-Government publications . The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract. (see 6.2).

ASTM INTERNATIONAL

ASTM D4814	-	Standard Specification for Automotive Spark-Ignition Engine Fuel

(Copies of these documents are available from American Society for Testing and Materials, P.O. Box C700, West Conshohocken, PA 19428-2959 or website: http://www.astm.org/.)

GENERAL DYNAMICS LAND SYSTEMS (GDLS)

LS199000006173	-	Nuclear Hardening Criteria

(Copies of this document are available from www.gdls.com or General Dynamics Land Systems, 38500 Mound Road, Sterling Heights, MI 48310-3200.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENT

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials and manufacturing processes. Design, materials, and manufacturing process selection shall be specified herein and in applicable referenced specifications, standards, and drawings. The lead ignition shall be fabricated and assembled in accordance with Drawing 12286829. Materials shall be uniform and free from imperfections or defects which affect their performance and serviceability. External surfaces shall be corrosion resistant. The materials shall be capable of meeting all requirements specified herein (see 4.2.3).

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs (see 4.2.3).

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3.2.2 Wire. The ignition lead wire shall be number 16 American Wire Gage (AWG) minimum (see 4.2.3).

3.3 Operating requirements. The ignition lead shall provide the following functional, operational, and performance capabilities.

3.3.1 Nuclear control. The ignition lead shall have optimum protection against electromagnetic impulse (EMP) and transient radiation effects. There shall be at least 2 layers of copperweld shielding or an equivalent shielding material (see 4.2.4.1).

3.3.2 Insulation resistance. The ignition lead insulation resistance between conductor and the cable shield shall be a minimum of 100 megohms at 500 ± 10 volts (V) direct current (dc) (see 4.2.4.2).

3.3.3 Maximum voltage. The ignition lead insulation shall show no evidence of breakdown when subjected to 20 kilovolts (kV) root mean square (rms) voltage between the conductor and the cable shield for a 5 to 10 second duration (see 4.2.4.3).

3.3.4 Accelerated aging. The ignition lead insulation shall withstand a temperature of 392 degrees Fahrenheit (°F) (200 degrees Celsius (°C)) for a minimum of 168 hours and continue to meet the requirements of 3.3.2 and 3.3.3 without degradation in performance (see 4.2.4.4).

3.3.5 Nuclear hardening. The ignition lead shall withstand a nuclear threat environment as defined in “Nuclear Hardening Criteria,” document control number LS199000006173 (see 4.2.4.5).

3.4 Interface requirements. The lead ignition shall accommodate the following inputs and interfaces.

3.4.1 Electromagnetic interference (EMI). The ignition lead shall meet the requirements of MIL-STD-461, or an equivalent, on a component level as part of the engine electronic control system (see 4.2.5.1).

3.5 Support and ownership requirements. The ignition lead shall meet the following logistics requirements.

3.5.1 Identification marking. The ignition lead shall be permanently marked and identified in accordance with Drawing 12286829 and MIL-STD-130, or equivalent (see 4.2.6.1).

3.5.2 Interchangeability. The ignition lead shall be interchangeable with other parts having identical part number (see 4.2.6.2).

3.5.3 Safety. The lead ignition shall pose no hazards, physically or electrically, to operating personnel (see 4.2.6.3).

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3.6 Operating environment requirements. The ignition lead shall operate under the following environmental conditions without damage of materials or degradation of performance.

3.6.1 Temperature stability. The ignition lead shall meet the requirements of 3.3.2 and 3.3.3 after exposure to a temperature range of –65 °F (–54 °C), 77 °F (25 °C), and 212 °F (100 °C) (see 4.2.7.1 through 4.2.7.3).

3.6.2 Operational altitude range. The ignition lead shall meet the requirements of 3.3.2 and 3.3.3 within a range of altitude between 1300±5 percent (%) feet below sea level and 10 000 ±2% feet above sea level (see 4.2.7.4).

3.6.3 Humidity. The ignition lead shall meet the requirements of 3.3.2 and 3.3.3 during and after being subjected to a relative humidity of 0 to 100% for a 15 day period. Upon completion, there shall be no corrosion or degradation in the strength of the lead (see 4.2.7.5).

3.6.4 Salt spray. The ignition lead shall meet the requirements of 3.3.2 and 3.3.3 after being exposed to a salt fog (5% NaC1 and 95% distilled water by weight) at 90 to 95 °F (32 to 35 °C) for a minimum of 48 hours (see 4.2.7.6).

3.6.5 Sand and dust. The ignition lead insulation resistance and maximum voltage performance shall not be deteriorated by exposure to sand and dust (see 4.2.7.7).

3.6.6 Fungus. The ignition lead insulation resistance and maximum voltage performance shall not be deteriorated when exposed to a fungus environment (see 4.2.7.8).

3.6.7 Chemical resistance. The ignition lead shall meet the requirements of 3.3.2 and 3.3.3 when exposed to the following materials in operating, non-operating and transit conditions (see 4.2.7.9):

a. Fuel vapors per A-A-52557, ASTM D4814, MIL-DTL-5624, and MIL-DTL-16884, or equivalents.
b. Hydraulic fluid per MIL-PRF-6083, or an equivalent.
c. Lubricating oil per MIL-PRF-7808 and MIL-PRF-23699, or equivalents.

3.6.8 Vibration. The ignition lead shall meet the requirements of 3.3.2 and 3.3.3 after being exposed to vibration levels up to 5 gravity units (g) from 5 to 2000 to 5 hertz (Hz) for a period of 20 minutes (see 4.2.7.10).

3.6.9 Shock.

3.6.9.1 Gun firing shock. The ignition lead shall be capable of sustaining shock impulses of 100±10 g intensity for a 1.5±0.2 milliseconds (ms) duration in any direction without malfunction (see 4.2.7.11.1).

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3.6.9.2 Ballistic shock. The ignition lead shall be capable of sustaining impact shock waves generated by the application of 200±20 g half-sine wave shock pulses in any direction without malfunction (see 4.2.7.11.2).

3.6.9.3 Basic shock. The ignition lead shall be capable of sustaining half-sine wave shock pulses of 40 g intensity for a 18 ms duration in any direction without malfunction (see 4.2.7.11.3).

3.6.10 Submergence. The ignition lead shall be fully operational after submergence in 6 feet (ft) of water for a period of 30 minutes (see 4.2.7.12).

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspections (CI) (see 4.1.2).

4.1.1 First article inspection. First article inspection shall be performed on 3 samples of the initial production ready production-representative units of an order or when required by the contract. When first article inspection is required (see 3.1, 6.2 and 6.3), it includes all verifications listed in table I.

4.1.2 Conformance inspection. Conformance inspection shall include the examinations and tests specified in table I.

4.1.2.1 Sampling plan. The sampling plan shall be as specified in the contract or order (see 6.2).

4.1.2.2 Examination. The samples selected in accordance with 4.1.2.1 shall be examined as specified in table I. The acceptance number in all cases is zero.

4.1.2.3 Tests (100% inspection). Each ignition lead shall be subjected to and pass the acceptance tests specified in table I.

4.2 Verification methods. Acceptable verification methods included in this section are visual inspection, measurements, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or

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cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification (see 6.2).

Table I. Verification methods.

Title	Requirement	Verification	First article	CI	
				Examination	Test
Design, materials and manufacturing	3.2 and 3.2.2	4.2.3	X	X	
Operating requirements	3.3	4.2.4			
Nuclear control	3.3.1	4.2.4.1	X		
Insulation resistance	3.3.2	4.2.4.2	X		X
Maximum voltage	3.3.3	4.2.4.3	X		X
Accelerated aging	3.3.4	4.2.4.4	X		
Nuclear hardening	3.3.5	4.2.4.5	X		
Interface requirements	3.4	4.2.5			
Electromagnetic interference	3.4.1	4.2.5.1	X		
Support and ownership requirements	3.5	4.2.6			
Identification and marking	3.5.1	4.2.6.1	X	X	
Interchangeability	3.5.2	4.2.6.2	X	X	
Safety	3.5.3	4.2.6.3	X	X	
Operating environment requirements	3.6	4.2.7			
High temperature	3.6.1	4.2.7.1.1	X		
Room temperature	3.6.1	4.2.7.1.2	X		
Low temperature	3.6.1	4.2.7.1.3	X		
Operational altitude range	3.6.2	4.2.7.4	X		
Humidity	3.6.3	4.2.7.5	X		
Salt spray	3.6.4	4.2.7.6	X		
Sand and dust	3.6.5	4.2.7.7	X		
Fungus	3.6.6	4.2.7.8	X		
Chemical resistance	3.6.7	4.2.7.9	X		
Vibration	3.6.8	4.2.7.10	X		
Shock	3.6.9	4.2.7.11	X		
Gun firing shock	3.6.9.1	4.2.7.11.1	X		
Ballistic shock	3.6.9.2	4.2.7.11.2	X		
Basic shock	3.6.9.3	4.2.7.11.3	X		
Submergence	3.6.10	4.2.7.12	X		

4.2.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be conducted at an atmospheric pressure of approximately 30 inches of mercury and at a temperature of 70 °F to 80 °F (21 °C to 27 °C) and a relative humidity of 90% or less.

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4.2.2.1 Inspection tolerances. Unless otherwise specified herein, all inspection conditions, as indicated by instrumentation, shall not deviate from the conditions specified herein by more than stated below.

a. Temperature: Under +100 °F (38 °C) ±4 °F (2.2 °C)
   Over +100 °F (38 °C) ±4 °F (2.2 °C)
b. Pressure: ±5%
c. Humidity: ±5% relative
d. Vibration amplitude:
   (1) Random: ±3 decibels at resonance and ±1 decibel elsewhere
   (2) Sinusoidal: ±2%
e. Vibration frequency: + 2%
f. Shock: + 5%
g. Acceleration: + 5%
h. Voltage, current and time:
   (1) Value specified as a whole number ± 0.5
   (2) Value specified to 1 decimal place ± 0.05
   (3) Value specified to 2 decimal place ± 0.005

4.2.3 Design, materials, and manufacturing. Conformance to 3.2 through 3.2.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to the requirements specified herein. Applicable records shall include drawing, specification, design data, receiving inspection records, processing quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.2.4 Operating requirements verification.

4.2.4.1 Nuclear control. To determine conformance to 3.3.1 the presence of at least two layers of copperweld shielding or equivalent shielding material shall be verified.

4.2.4.2 Insulation resistance test. To determine conformance to 3.3.2, measure the insulation resistance between conductor and cable shield using an applied potential of 490 to 510 Vdc. The resistance shall be less than 100 megohms

4.2.4.3 Maximum voltage test. To determine conformance to 3.3.3, a voltage of 20 kV rms shall be applied between the conductor and the cable shield for 5 to 10 seconds without evidence of insulation breakdown.

4.2.4.4 Accelerated aging test. To determine conformance to 3.3.4, the ignition lead shall be placed dry, in an air oven and maintained at a maximum operating ambient temperature of 392°F (200°C) for a minimum period of 168 hours. Upon completion of the 168 hours, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

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4.2.4.5 Nuclear hardening. To determine conformance to 3.3.5, verify that the ignition lead meets nuclear environment requirements as specified in LS199000006173, shall be determined by the acquisition activity on the basis of circuit analysis and laboratory testing or through nuclear test of a complete vehicle.

4.2.5 Interface requirements verification.

4.2.5.1 Electromagnetic interference (EMI). To determine conformance to 3.4.1, the ignition lead shall be tested in accordance with MIL-STD-461 or equivalent.

4.2.6 Support and ownership requirements verification.

4.2.6.1 Identification marking. To determine conformance to 3.5.1, verify the presence of the required markings on the lead ignition. After performing all environmental tests in 4.2.7, re-inspect the lead ignition markings for readability.

4.2.6.2 Interchangeability. To determine conformance to 3.5.2, the interchangeability between different parts, subassemblies, and assemblies having the same military part number shall be verified using one or more of the methods described in 4.2.

4.2.6.3 Safety. To determine conformance to 3.5.3, the lead ignition shall be inspected to assure that the mechanical components are free of hazardous burrs, nicks, sharp edges, foreign materials, or other imperfections posing physical danger to the operators, and that the lead ignition electrical components are free of exposed, frayed, unsecured, or otherwise improperly protected circuits posing electrical danger to the operators.

4.2.7 Operating environment requirements verification.

4.2.7.1 High temperature test. To determine conformance to the high temperature portion of 3.6.1 and following test 4.2.4.4, the ignition lead shall be subjected to a high temperature of 212°F (100°C) for a period of 100 hours. Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.2 Room temperature test. To determine conformance to the room temperature portion of 3.6.1 and following test 4.2.7.1, the ignition lead shall be subjected to room a temperature of 77°F (25°C) for a period of 8 hours. Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.3 Low temperature test. To determine conformance to 3.6.1 and following test 4.2.7.2, the ignition lead shall be subjected to low temperature of –65°F (–54°C) for a period of 30 hours. Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.4 Operational altitude range test. To determine conformance to 3.6.2, ignition lead shall be subjected to a maximum atmospheric pressure of 31.19 inches of mercury (1300 feet

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below sea level) and a minimum atmospheric pressure of 20.01 inches of mercury (10,000 feet above sea level). Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.5 Humidity test. To determine conformance to 3.6.3, the ignition lead shall be subjected to an atmosphere of clear water vapor for a period of 15 days under the conditions and cycle specified in table II. Immediately after 12 hours of the first test cycle, check the ignition lead for proper operation with the maximum specified voltage. During the last cycle of the 15 day test, operate the ignition lead throughout its entire range of operation under each of the ambient conditions specified. At the end of the 15 days of test, dry the ignition lead without forced convection for 12 hours in air at less than 69.8°F (21°C) with not less than 50% relative humidity. After this, disassemble and inspect the ignition lead. There shall be no corrosion or other defects present that affect the function or structural strength of the assembly. Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

Table II. Humidity conditions test cycle.

Time period (hours)	Relative humidity (%)	Ambient temperature
8	95 ± 5	129 ± 5.4°F (54 ± 3°C)
4	Holding chamber moisture content constant	129 ± 5.4°F (54 ± 3°C) gradually down to 70 ± 5.4°F (21 ± 3°C)
8	Holding chamber moisture content constant	70 ± 5.4°F (21 ± 3°C)
4	Up to 95 ± 5, holding chamber moisture content or adding vapor to reach required value	70 ± 5.4°F (21 ± 3°C) Uniformly up to 129 ± 5.4°F (54 ± 3°C)

4.2.7.6 Salt spray. To determine conformance to 3.6.4, ignition lead shall be exposed to salt fog for a minimum of 48 hours. The fog solution shall be 5 % NaCl by weight in 95 % distilled water by weight. The temperature shall be maintained between 90 to 95°F (32 and 35°C). Fog density shall be 3 quarts of solution per 10 cubic feet of chamber volume per 24 hours. Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.7 Sand and dust test. To determine conformance to 3.6.5, the ignition lead shall be subjected to the sand and dust test in accordance with Method 510 of MIL-STD-810 or an equivalent. The ignition lead need not be operated during the test; however, at the conclusion of the test and prior to any cleaning of the ignition lead, it shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.8 Fungus test. To determine conformance to 3.6.6, ignition lead shall be inoculated with a spore suspension and tested in accordance with Method 508 of MIL-STD-810 or an equivalent. Temperature shall range from 80 to 84°F and the relative humidity from 96 to 100 %

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for a period of 28 days. Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.9 Chemical test. To determine conformance to 3.6.7, ignition lead shall be exposed to fluids and vapors of materials listed in 3.6.7. Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.10 Vibration test. To determine conformance to 3.6.8, ignition lead shall be subjected to vibrations when normally mounted (with vibration isolators in place) and powered. The vibrations shall be of 5 g magnitude in each direction along 3 mutually perpendicular axes at a logarithmic sweep rate. The vibration cycle time shall be equivalent to 20 minutes per axis from 5 to 2000 to 5 Hertz (Hz) with each 20 minute test period representing 1000 miles of vehicle operation. Conduct a resonance sweep to identify up to four of the most severe resonance and then accomplish resonance dwells for one-sixth of the total cycling vibration time at each resonance at the specified levels. Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.11 Shock.

4.2.7.11.1 Gun firing shock test. To determine conformance to 3.6.9.1, the ignition lead while mounted normally (with vibration isolators in place, if any) and powered shall be subjected to 3 shocks of 100 $\pm$ 10 g, 1.5 $\pm$ 0.2 milliseconds (ms) along each of 3 mutually perpendicular axes (18 shocks). Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.11.2 Ballistic impact shock test. To determine conformance to 3.6.9.2, the ignition lead while mounted normally (with vibration isolators in place, if any) and powered shall be subjected to 3 shock waves of 200 $\pm$ 20 g half sine shock pulses along each of 3 mutually perpendicular axes (18 shocks). Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.11.3 Basic shock test. To determine conformance to 3.6.9.3, the ignition lead while mounted normally (with vibration isolators in place, if any) and powered shall be subjected to 3 half-sine wave shock pulses of 40 g, 18 ms along each of 3 mutually perpendicular axes (18 shocks). Upon completion, the lead shall be subjected to and pass the tests of 4.2.4.2 and 4.2.4.3.

4.2.7.12 Submergence test. To determine conformance to 3.6.10, ignition lead shall be tested in association with a fording test of a vehicle. The ignition lead when installed on an engine shall be utilized in the starting system. It shall be capable of meeting all performance requirements after submergence in 6 feet of water for 30 minutes.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The ignition lead covered by this specification is intended to provide electrical continuity between the ignition exciter and spark igniter on the AGT 1500 gas turbine engine in the M1 (Abrams) military tank. This ignition lead is military unique in application and design, due to the nature of the environmental extremes it should function under.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. If required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).
c. When first article is required (see 3.1 and 6.3).
d. If first article sample size is other than as specified (see 4.1.1).
e. Sampling plan (see 4.1.2.1).
f. If verification alternatives are other than specified (see 4.2.1)
g. Packaging requirements (see 5.1).

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors (see 6.2). This guidance should cover whether the first article is a first article sample, a first article production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing should furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

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6.4 Subject term (key word) listing.

Electrical starter
Engine exciters
High energy
High tension voltage

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2920-2010-004)

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